# Terraform Assignment

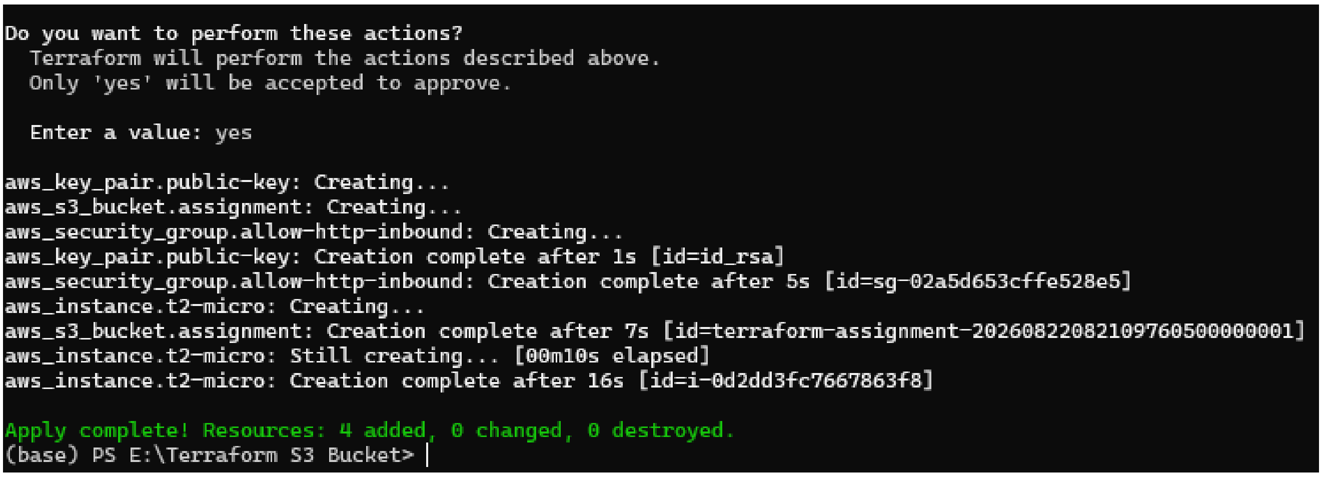
EC2 and S3 Resource Lifecycle

---

Provisioning and cleanup of AWS resources using Terraform

This document demonstrates provisioning a **t2.micro EC2 instance** and an **S3 bucket** in the **us-east-1** region using Terraform, validating both in the AWS Console, and destroying all resources with `terraform destroy`. Both resources are tagged with `Name = "TerraformAssignment"`.

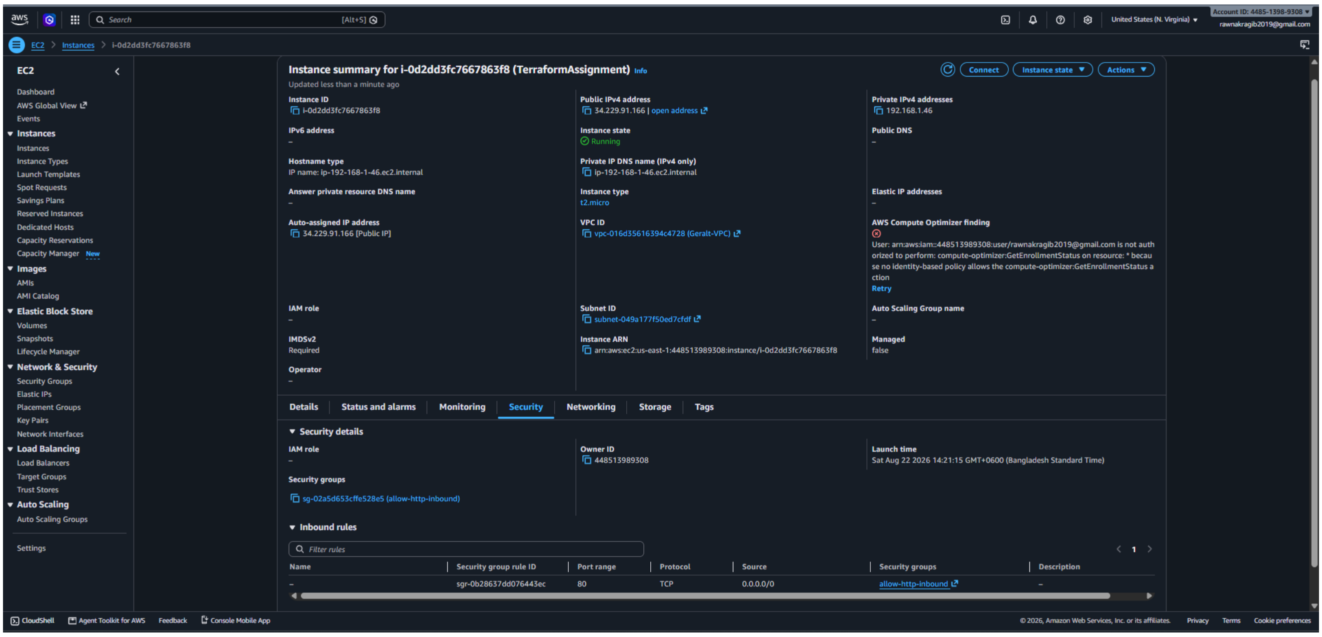
## 1. terraform apply — Execution



```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_key_pair.public-key: Creating...
aws_s3_bucket.assignment: Creating...
aws_security_group.allow-http-inbound: Creating...
aws_key_pair.public-key: Creation complete after 1s [id=id_rsa]
aws_security_group.allow-http-inbound: Creation complete after 5s [id=sg-02a5d653cffe528e5]
aws_instance.t2-micro: Creating...
aws_s3_bucket.assignment: Creation complete after 7s [id=terraform-assignment-20260822082109760500000001]
aws_instance.t2-micro: Still creating... [00m10s elapsed]
aws_instance.t2-micro: Creation complete after 16s [id=i-0d2dd3fc7667863f8]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.
(base) PS E:\Terraform S3 Bucket>
```

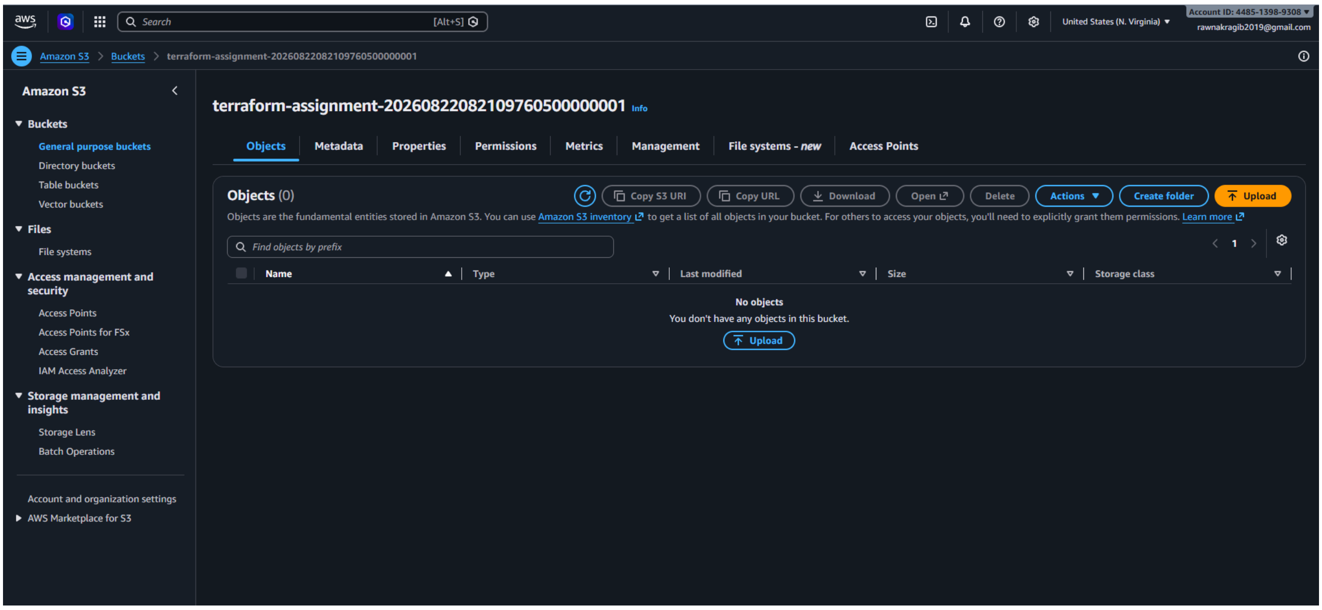
Terminal output of `terraform apply`. **Apply complete! Resources: 4 added, 0 changed, 0 destroyed.**

## 2. EC2 Instance in AWS Console



Instance `i-0d2dd3fc7667863f8` (TerraformAssignment) — type **t2.micro**, state **Running**, in region N. Virginia (us-east-1).

# 3. S3 Bucket in AWS Console



Bucket `terraform-assignment-20260822082109760500000001` created with a unique auto-generated name.

## 4. terraform destroy — Successful Cleanup

```
Plan: 0 to add, 0 to change, 10 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

aws_route_table_association.public-rt-assoc: Destroying... [id=rtbassoc-023574db121206e8c]
aws_route.default-route: Destroying... [id=r-rtb-0cfd78607e28dd3201080289494]
aws_s3_bucket.assignment: Destroying... [id=terraform-assignment-20260822082109760500000001]
aws_instance.t2-micro: Destroying... [id=i-0d2dd3fc7667863f8]
aws_s3_bucket.assignment: Destruction complete after 1s
aws_route_table_association.public-rt-assoc: Destruction complete after 1s
aws_route.default-route: Destruction complete after 2s
aws_internet_gateway.internet-access: Destroying... [id=igw-085c3ee8b59201ebc]
aws_route_table.public-rt: Destroying... [id=rtb-0cfd78607e28dd320]
aws_route_table.public-rt: Destruction complete after 2s
aws_instance.t2-micro: Still destroying... [id=i-0d2dd3fc7667863f8, 00m10s elapsed]
aws_internet_gateway.internet-access: Still destroying... [id=igw-085c3ee8b59201ebc, 00m10s elapsed]
aws_instance.t2-micro: Still destroying... [id=i-0d2dd3fc7667863f8, 00m20s elapsed]
aws_internet_gateway.internet-access: Still destroying... [id=igw-085c3ee8b59201ebc, 00m20s elapsed]
aws_internet_gateway.internet-access: Destruction complete after 21s
aws_instance.t2-micro: Destruction complete after 23s
aws_key_pair.public-key: Destroying... [id=id_rsa]
aws_subnet.public: Destroying... [id=subnet-049a177f50ed7cfdf]
aws_security_group.allow-http-inbound: Destroying... [id=sg-02a5d653cffe528e5]
aws_key_pair.public-key: Destruction complete after 1s
aws_subnet.public: Destruction complete after 2s
aws_security_group.allow-http-inbound: Destruction complete after 2s
aws_vpc.Geralt: Destroying... [id=vpc-016d35616394c4728]
aws_vpc.Geralt: Destruction complete after 2s

Destroy complete! Resources: 10 destroyed.
(base) PS E:\Terraform S3 Bucket>
```

Terminal output of `terraform destroy`. **Destroy complete! Resources: 10 destroyed.**

# 5. Terraform Configuration (main.tf)

```
terraform {
    required_providers {
    aws = {
        source  = "hashicorp/aws"
        version = "~> 5.7.0"
    }
}

}

provider "aws" {
    profile = "Geralt"
    region = "us-east-1"

}

data "aws_ami" "amazon_linux" {
    most_recent = true
    owners      = ["amazon"]

    filter {
        name   = "name"
        values = ["al2023-ami-*-x86_64"]
    }

    filter {
        name   = "virtualization-type"
        values = ["hvm"]
    }
}

resource "aws_vpc" "Geralt" {
    cidr_block = "192.168.0.0/16"
    tags = {
        Name = "Geralt-VPC"
    }

}

resource "aws_subnet" "public" {
    vpc_id = aws_vpc.Geralt.id
    cidr_block = "192.168.1.0/24"
    availability_zone = "us-east-1a"
}

resource "aws_route_table" "public-rt" {
    vpc_id = aws_vpc.Geralt.id
    tags = {
        Name = "public-route-table"
    }
}

resource "aws_internet_gateway" "internet-access" {
    vpc_id = aws_vpc.Geralt.id
    tags = {
        Name = "main-igw"
    }
}

resource "aws_route" "default-route" {
    route_table_id = aws_route_table.public-rt.id
    destination_cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.internet-access.id

}

resource "aws_route_table_association" "public-rt-assoc" {
  subnet_id = aws_subnet.public.id
  route_table_id = aws_route_table.public-rt.id
}

resource "aws_security_group" "allow-http-inbound" {
    name = "allow-http-inbound"
    description = "sg for allowing inbound traffic"
    vpc_id = aws_vpc.Geralt.id
```

```
    ingress {
        from_port = 80
        to_port = 80
        protocol = "tcp"
        cidr_blocks = ["0.0.0.0/0"]
    }
    egress {
        from_port = 0
        to_port = 0
        protocol = "-1"
        cidr_blocks = ["0.0.0.0/0"]
    }

}

resource "aws_key_pair" "public-key" {
    key_name = "id_rsa"
    public_key = file("C:/Users/User/.ssh/id_rsa.pub")
}

resource "aws_instance" "t2-micro" {
    ami = data.aws_ami.amazon_linux.id
    instance_type = "t2.micro"
    key_name = aws_key_pair.public-key.key_name
    subnet_id = aws_subnet.public.id
    vpc_security_group_ids = [ aws_security_group.allow-http-inbound.id ]
    associate_public_ip_address = true
    tags =  {
        Name = "TerraformAssignment"
    }

}

resource "aws_s3_bucket" "assignment" {
    bucket_prefix = "terraform-assignment-"
    tags = {
        Name = "TerraformAssignment"
    }
}
```